

The Effect Of Extreme Climate Events On Migration In Mexico

Luca Veh



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- According to the Internal Displacement Monitoring Centre (IDMC) in 2025 the number of internally displaced people reached 84 million worldwide - the highest figure ever recorded and equivalent to the entire population of Germany.
- The amount has doubled since 2018.
- Climate related disasters are one of the main drivers of this displacement.
- Increasing our understanding of how people are displaced and forced to migrate improves our ability to react and provide targeted and effective aid.

- Mexico is among the countries most exposed to extreme climate events in the Americas
- The National Centre for Disaster Prevention (CENAPRED) records an average of 325.1 severe extreme weather events per year.
- For Mexico the IDMC reports a median amount of 53,351 displaced people per year, with some years peaking at over 800,000 displaced people per year.

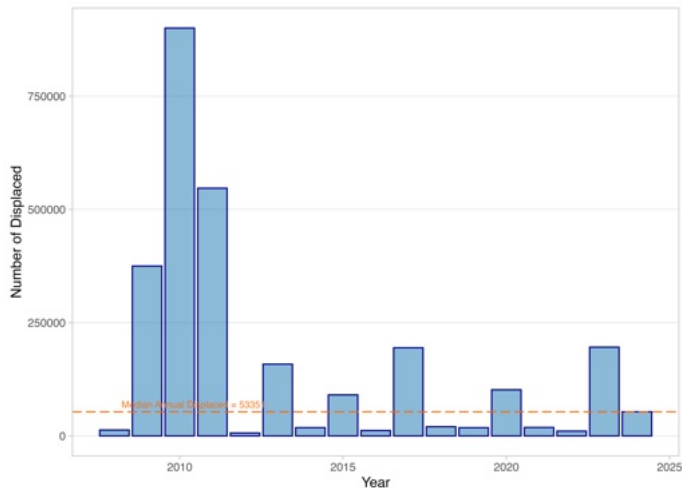


Figure: Number of Displaced People per Year

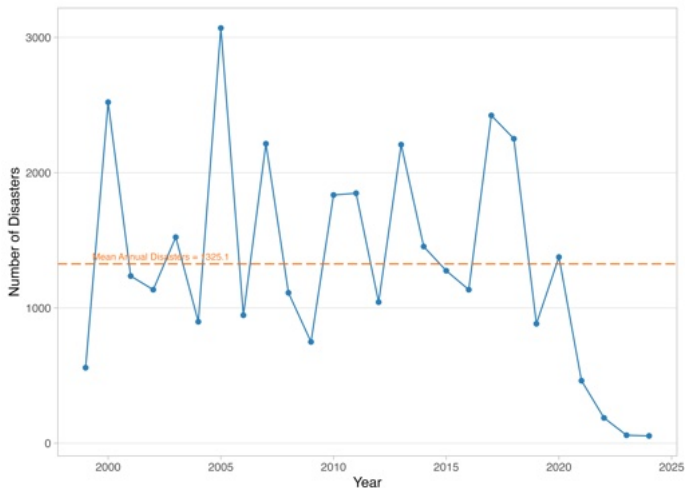
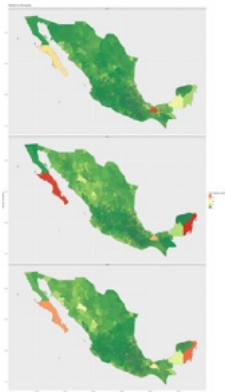


Figure: Number of Disasters per Year

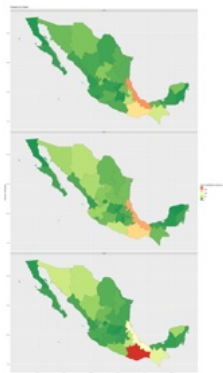
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How do extreme weather events influence medium-term migration patterns in Mexico?

- Extreme weather events include tropical storms, floods, heats, droughts, etc. as defined by the Atlas Index from CENAPRED
- Medium-term: 5-year intervals (following the census)
- Internal migration: Only migration inside of Mexico, specifically between mexican municipalities.



(a) Disasters in Municipalities



(b) Disasters in States

Figure: Spatial distribution of disasters at different administrative levels

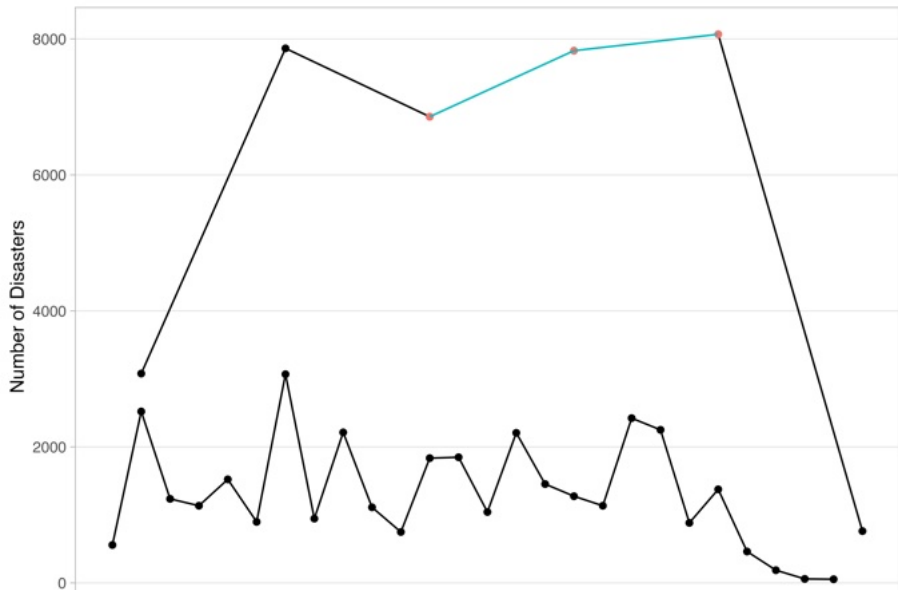
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- Migration data comes from the Mexican Population and Housing Census
- Collected every five years - relevant years for this paper are the 2010, 2015 and 2020 iterations
- In each census round, people are asked where they resided five years ago.
- The migration outcomes are then calculated as the amount of people who moved in (current municipality of residence) and the amount of people who moved out (municipality of residence five years ago) for each municipality.

- The Atlas Nacional de Riesgos is a data base that contains municipality level information about different kind of extreme weather events in Mexico
- To be added to this data base, local officials need to classify an extreme weather event as an emergency that would overwhelm local resources.
- This is thus not raw weather data, but a selection of events that were considered by local authorities to have a substantial impact on their municipality.
- The selection of events might also reflect local capacity to adequately report events and request federal aid.

- After pre-processing, the datasets for migration, disasters, demographics and crime were merged into a municipality-node dataset.
- All data was aggregated to the five year census intervals.
- The municipality-node dataset contains one entry for each node (=municipality) and census year.

Disaster Aggregation Plot



Summary statistics (municipality-year level)

Variable	N	Mean (SD)	Median	Min	Max
Panel A: Migration outcomes					
Period Immigration	6975	2512.33 (11059.83)	344.00	0.00	276699.00
Period Emigration	6975	2512.33 (13784.52)	386.00	0.00	483051.00
Net Immigration	6975	0.00 (8410.35)	-4.00	-219655.00	113326.00
Immigration Rate	6975	0.04 (0.03)	0.03	0.00	0.67
Emigration Rate	6975	0.04 (0.05)	0.03	0.00	2.10
Net Immigration Rate	6975	-0.00 (0.06)	-0.00	-2.06	0.66
Panel B: Disaster exposure					
Total Unique Disasters (Period)	6975	3.79 (4.79)	2.00	0.00	86.00
Total Disasters (Period)	6975	3.26 (4.02)	2.00	0.00	78.00
Panel C: Controls					
Mean Income	6965	3313.14 (2378.09)	2941.76	34.95	39125.43
Mean Age	6975	30.99 (3.71)	30.66	20.24	48.91
Female Rate	6975	0.51 (0.02)	0.51	0.43	0.61
Literacy Rate	6966	0.86 (0.08)	0.88	0.42	1.00
Mean Years of Schooling	6966	6.08 (1.28)	6.01	2.07	12.62
Municipal Population	6975	51188.19 (200573.91)	12704.00	76.00	5413803.00
Population Density	6975	219.80 (719.69)	51.75	0.11	10714.21
Crime (Homicides)	6975	50.18 (411.69)	1.00	0.00	15140.00

Notes: Unit of observation is municipality-year. N reports non-missing observations.

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- To estimate the medium-run effect of extreme weather events on migration outcomes the identification strategy relies on the following TWFE model:

$$NetMigration_{it} = \beta ClimateShock_{it} + \gamma X_{it} + \mu_i + \rho_t + \epsilon_{it}$$

where:

- $ClimateShock_{it}$: continuous climate shock variable
- X_{it} : set of controls
- μ_i : Municipality Fixed Effects
- ρ_t : Period Fixed Effects

- For the TWFE model with a continuous shock variable to estimate the causal impact of extreme weather events on migration outcomes Strict Exogeneity is required:

$$E[\epsilon_{it} | ClimateShock_{i,t}, X_{it}, \mu_i, \rho_t] = 0 \quad \forall t \in 2010, 2015, 2020$$

- In words of the parallel trends assumption this also implies that municipalities, that experienced more or fewer extreme weather events in a given year, relative to their average, would have had the similar migration outcomes as other municipalities, in the absence of climate exposure, conditional on FEs and controls.

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Table 2: Main Results

Dependent Variable:	Net Immigration			
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Constant	-115.2 (360.6)			
Total disasters (period)	35.31 (111.8)	4.603 (27.67)	-29.21 (28.82)	-40.61 (36.09)
Municipal population			0.0486 (0.0339)	0.0492 (0.0342)
Mean income			0.2432** (0.0990)	0.2473** (0.0997)
Average age			-182.4* (95.40)	-178.1* (96.06)
Female share			-13,188.3*** (4,635.0)	-12,918.0*** (4,555.0)
Literacy rate			7,561.0*** (1,932.7)	7,324.0*** (1,903.5)
Urbanization rate			512.7 (748.3)	512.1 (743.3)
Crime rate (homicides)			-0.6376 (1.091)	-0.6298 (1.096)
Total disasters (lag 1)				-40.18 (59.11)
<i>Fixed-effects</i>				
Municipality FE		Yes	Yes	Yes
Year FE		Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	6,975	6,975	6,965	6,965
R ²	0.00028	0.84391	0.84883	0.84892

Clustered (Municipality FE) standard-errors in parentheses

Signif. Codes: ***, 0.01, **, 0.05, *, 0.1

Model 1:

- A simple pooled OLS regression
- Coefficient has different sign than Model 3

Model 2:

- Includes both municipality and year fixed effects but no control variables.
- Avoid bias from controlling with variables that are themselves part of a potential causal pathway through which disasters induce migration.
- Likely suffers from OVB.

Model 3:

- Main regression specification that includes both fixed effects, as well as a full set of controls.

Model 4:

- Includes disaster coefficient lagged by one (5-year) time period.
- Interpret as lagged response to disasters that happened in the previous 5-year period.
- Disaster coefficient is not statistically significant.

Results for Model 3:

- Overall average ceteris paribus effect of an additional disaster on net migration, using only within-municipality, over-time derivations, net of common year shocks and conditional on control variables is -29.21.
- On average c.p. an additional disaster is associated with 29.21 more people leaving a municipality than enter it. The effect is not statistically significant.

- The coefficient of the disaster coefficient remains not statistically significant across specifications
- This implies that on average there is no direct medium-term effect of extreme weather events, as measured by the Atlas Declaratorias, on net migration in Mexican municipalities.
- Standard errors are also very high across specifications so all coefficients should be interpreted with caution.

Thank you!
Questions?